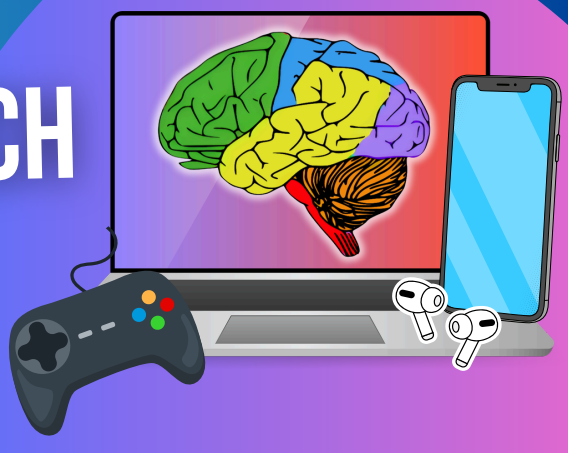
CENTNER ACADEMY

THE BRAIN SCIENCE OF TECH

PROJECT PLANNER

NEURAL NETWORKS: THE SCIENCE OF DOPAMINE & TECH



Welcome to the Neural Networks: The Science of Dopamine & Tech project!

As a middle schooler, you are currently living through the most significant biological "software update" of your life. Your brain is a masterpiece of engineering, but it’s also being targeted by the world’s most powerful algorithms. Every notification, "like," and infinite scroll is a calculated attempt to tap into your brain's reward system—but there is a massive difference between the "Cheap Dopamine" that leaves you feeling drained and the "Earned Dopamine" that comes from building, creating, and solving problems.

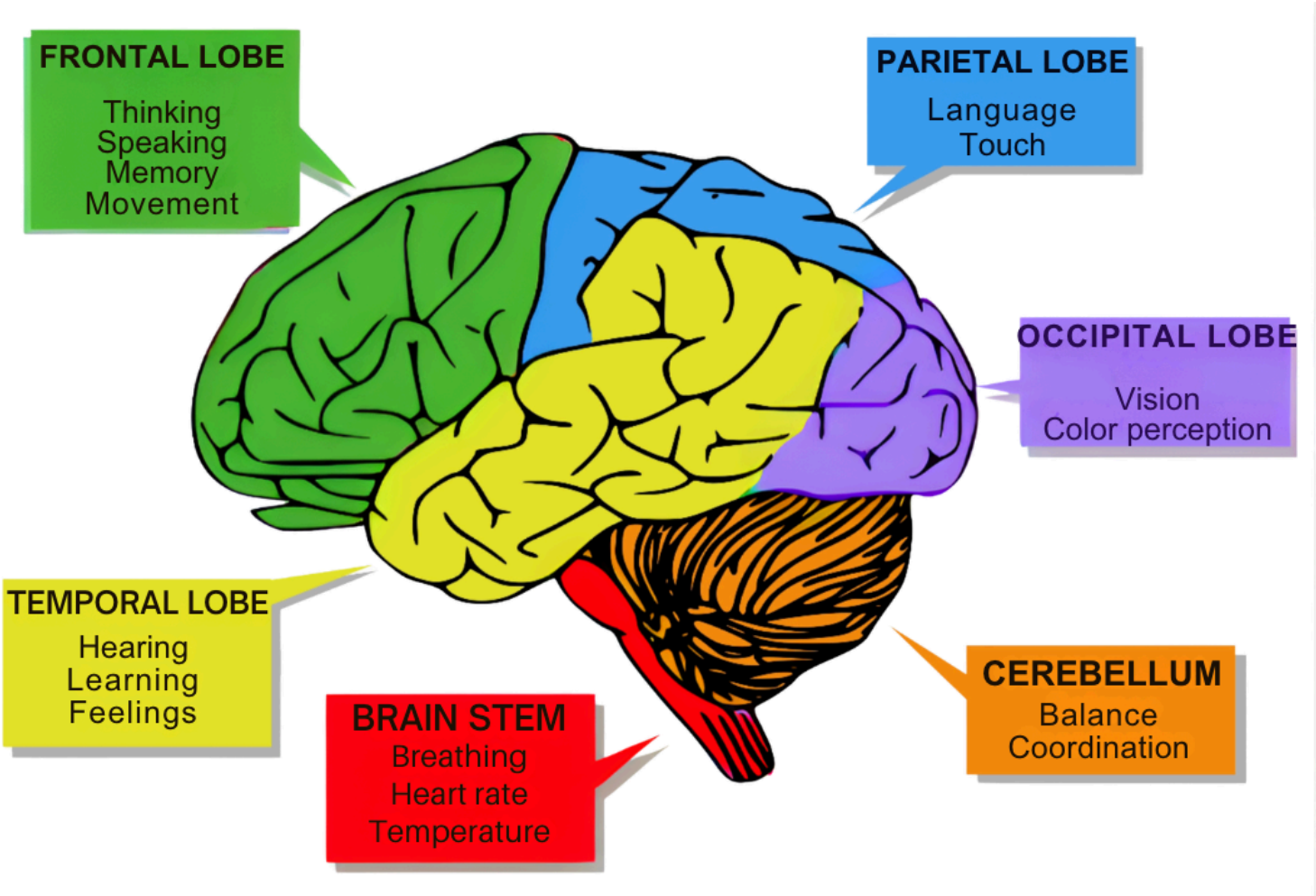
In this project, you aren't just a consumer; you are a Neural Architect. You will dive deep into a specific part of your brain to discover how it reacts to the technology in your pocket. You’ll perform a "Neural Audit" on your own habits, build an interactive digital experience to teach others, and finally, collaborate with your peers to construct a massive, 3D model of the human brain.

It’s time to move from being "programmed" by your apps to programming your own future. Let’s get to work!

Phase 1: Identify Your Neural Territory

Every part of your brain has a "day job."

***Instructions**: Below the following diagram, read the "Parts of our brain descriptions" carefully to better understand all brain territories. You will be assigned a part of the brain for your project to investigate to see how technology is changing and challenging its workload.*



Parts of our brain descriptions:

1. The Cerebrum (The "Executive Suite" & Sensory Hub)
The largest part of your brain, divided into four specialized lobes:

- **Frontal Lobe — *The CEO (Chief Executive Officer)*:**
  Known as the center for **decision-making, impulse control, and focus.**

Tech Connection: This is the part of you that tries to say "no" to one more YouTube video.

- **Parietal Lobe — *The Sensory Map*:**
  Known for **spatial awareness, navigation, and touch.**
  Tech Connection: This lobe helps you coordinate your thumbs during a fast-paced game or navigate a 3D virtual world.

- **Temporal Lobe — *The Sound & Memory Library*:**
  Known for **processing audio, language, and long-term memories.**
  Tech Connection: This area is constantly processing notification pings and viral audio loops.

- **Occipital Lobe — *The Visual Processor*:**
  Known as the **eyes in the back of your head.** *Tech Connection:* This lobe is on overdrive when you are watching high-definition video or fast-moving graphics.

2. The Cerebellum — *The "Auto-Pilot" Coordinator:*
Located at the back of the brain, this is known for **balance, posture, and fine motor skills.**

- *Tech Connection:* This is where your "muscle memory" lives—like being able to type or use a controller without looking at your hands.

3. The Brain Stem — *The "Life Support" System:*
The base of the brain that connects to the spine; it is known for **controlling automatic functions like breathing, heart rate, and sleep cycles.**

- *Tech Connection:* Technology "hijacks" this area when blue light from screens tricks your brain stem into thinking it's daytime, ruining your sleep.

Student Workspace: Preliminary Research

Project Basics: My Neural Territory

My Brain Part: _____

Primary Function: _____

How Technology Affects It: _____

One "Cheap Dopamine" Trigger for this part: _____

One "Earned Dopamine" Activity to strengthen this part: _____

1. The Hook:

The brain part being researched is _____, also known as _____**[Nickname, e.g., The CEO/The Life Support System]**. Its main job in my daily life is to _____.

2. The Tech Connection

Most people don't realize it, but this part of the brain is heavily impacted by _____ (e.g., TikTok scrolling, gaming, blue light). When this technology is used this technology, my brain part feels _____.

3. Cheap vs. Earned Dopamine

- **The Hijack:** A 'Cheap Dopamine' trigger for this part of my brain is _____, which makes me feel _____.
- **The Build:** An 'Earned Dopamine' activity I can do to strengthen this part of my brain is _____, which results in _____.

4. The Interactive Plan

For the digital project, we are planning to build a **[Website / Infographic / Video]** that will show _____.

Brain Parts Information: Technology & The "AKA"

1

The Frontal Lobe (aka The CEO)

- **The Job:** Managing focus, planning for the future, and stopping impulsive behavior.
- **The Tech Impact:** When you get a notification and immediately click it, you are bypassing "The CEO."
- **Cheap vs. Earned:** "Cheap" dopamine comes from clicking "Next Episode" on autoplay; "Earned" dopamine comes from the struggle of learning a new coding language or finishing a difficult level you designed yourself.

2

The Parietal Lobe (aka The Sensory Map)

- **The Job:** Handling touch, pressure, and where your body is in space.
- **The Tech Impact:** This part is highly active during gaming. It helps you "feel" the virtual world through the controller or screen.
- **Cheap vs. Earned:** "Cheap" is the mindless tapping of a clicker game; "Earned" is developing the complex hand-eye coordination required to build a 3D model in a design app.

3

The Temporal Lobe (aka The Sound & Memory Library)

- **The Job:** Processing what you hear and storing memories.
- **The Tech Impact:** Apps use "earworms" (repetitive sounds) to make you remember them.
- **Cheap vs. Earned:** "Cheap" is the dopamine hit from a catchy, 15-second sound loop on social media; "Earned" is the satisfaction of composing your own digital track or recording a podcast script.

4

The Occipital Lobe (aka The Visual Processor)

- **The Job:** Decoding everything your eyes see—colors, shapes, and motion.
- **The Tech Impact:** Fast-paced video edits (cuts every 2-3 seconds) keep this lobe in a state of constant "startle," making it hard to look away.
- **Cheap vs. Earned:** "Cheap" is watching high-speed, flashy "satisfying" videos; "Earned" is carefully editing your own video or creating a balanced color palette for a website.

5

The Cerebellum (aka The Auto-Pilot)

- **The Job:** Muscle memory, balance, and fine-tuning movement.
- **The Tech Impact:** This is why you can "thumb-type" a text message without looking at the keyboard.
- **Cheap vs. Earned:** "Cheap" is the repetitive, low-skill motion of scrolling; "Earned" is practicing the precision required to play a digital instrument or navigate a complex flight simulator.

6

The Brain Stem (aka The Life Support System)

- **The Job:** Keeping you alive (breathing/heartbeat) and managing your "Internal Clock" (Circadian Rhythm).
- **The Tech Impact:** Blue light from screens can trick the Brain Stem into thinking the sun is up, stopping the production of sleep hormones.
- **Cheap vs. Earned:** "Cheap" is staying up late for "just one more" game (hijacking your survival systems); "Earned" is setting a tech-boundary that allows your brain to rest and recover.

The Dopamine Ledger

Identify your tech habits from the last 24 hours. Fill in the remaining 3 rows.

Activity	Cheap Dopamine (Passive)	Earned Dopamine (Creative)	Neural Impact (How do you feel after?)
Example: Scrolling TikTok	*X*		*Foggy, tired, wanting more.*
Example: Coding a Game		*X*	*Proud, tired but satisfied.*

Phase 2: Research & Digital Build

Your mission is to teach others how technology re-wires the part of the brain you are researching.

Requirements for your Website or Presentation:

_____ ***Anatomy Deep Dive:*** *Diagram of your brain part and its primary function.*

_____ ***The "Hijack" Explanation:*** *How does "Cheap Dopamine" (notifications, infinite scroll) affect this specific area?*

_____ ***The "Growth" Explanation:*** *How does "Earned Dopamine" (building, learning) strengthen this area?*

_____ ***The Interactive Artifact:*** *Embed one of the following to demonstrate your point:*

- *A **Scratch Game** simulating a dopamine loop.*
- *An **Interactive Infographic** (Canva) with clickable hotspots.*
- *A **Video Tutorial** or "Neural Audit" guide for peers.*

Phase 3: The Physical Connection

You will build a 3D brain model as a class. How does your digital work live in the physical world?

- **Connection Point:** Identify where your part sits in the large-scale model.
- **The "QR" Bridge:** Create a QR code that links your physical "brain station" to your digital website/presentation.
- **Visual Element:** Design one physical 3D element (using cardboard, wires, or lights) that represents the "Neural Pathway" of your chosen part.